GROUP WEEKLY REPORTS

Project Title: Why 2024 Flights Ran Late
Team: Team 10 – DATA-VIZ HACK-A-BIT
Team Members: Annalakshmi R, Ashvitha I, Dharshini P, Joshna Amirthaa Mariappan

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WEEK 5 – EXPLORATORY DATA ANALYSIS

During the fifth week, our team conducted Exploratory Data Analysis (EDA) to understand the characteristics, distributions, and patterns present in the cleaned flight dataset.

We began by examining the numerical variables using descriptive statistics such as mean, median, minimum, maximum, and standard deviation. These statistics helped us understand the overall behavior of flight delays and identify differences between average and typical delay values.

Our team focused particularly on departure delay (`dep_delay`) and arrival delay (`arr_delay`). We explored their distributions and examined how delays varied across different airlines, months, and days.

We performed univariate analysis to study individual variables and bivariate analysis to investigate relationships between two variables. We used visualizations such as histograms, box plots, bar charts, and scatter plots to present the results clearly.

Through this analysis, we observed that many flights departed or arrived slightly earlier than their scheduled timings, while a smaller number experienced extremely long delays. These extreme values influenced the mean delay values and created positively skewed distributions.

We also examined how delay patterns varied across airlines and time periods. The team discussed how these patterns could provide useful insights into operational performance and passenger experience.

By the end of the week, we had identified several meaningful trends in the dataset and developed a better understanding of the factors associated with flight delays. The findings from EDA formed the basis for further statistical analysis and visualization.